

Functions — Test (G10 / IB Criterion A)

Name: _____ Class: _____ Date: _____

Time: 60–75 minutes (teacher choice)

Calculator: Allowed

Instructions

- Answer all questions.
- Show full working where appropriate.
- Leave exact answers unless told to round.
- Use correct mathematical notation.
- Keep your work organized using the question numbers.

Quick formulas

- For $y = \sqrt{\text{expression}}$, require $\text{expression} \geq 0$
- For $y = \frac{1}{\text{expression}}$, require $\text{expression} \neq 0$
- For $y = \frac{1}{\sqrt{\text{expression}}}$, require $\text{expression} > 0$
- For $y = \log_a(\text{expression})$, require $\text{expression} > 0$
- A function has an inverse only if it is one-to-one on its stated domain

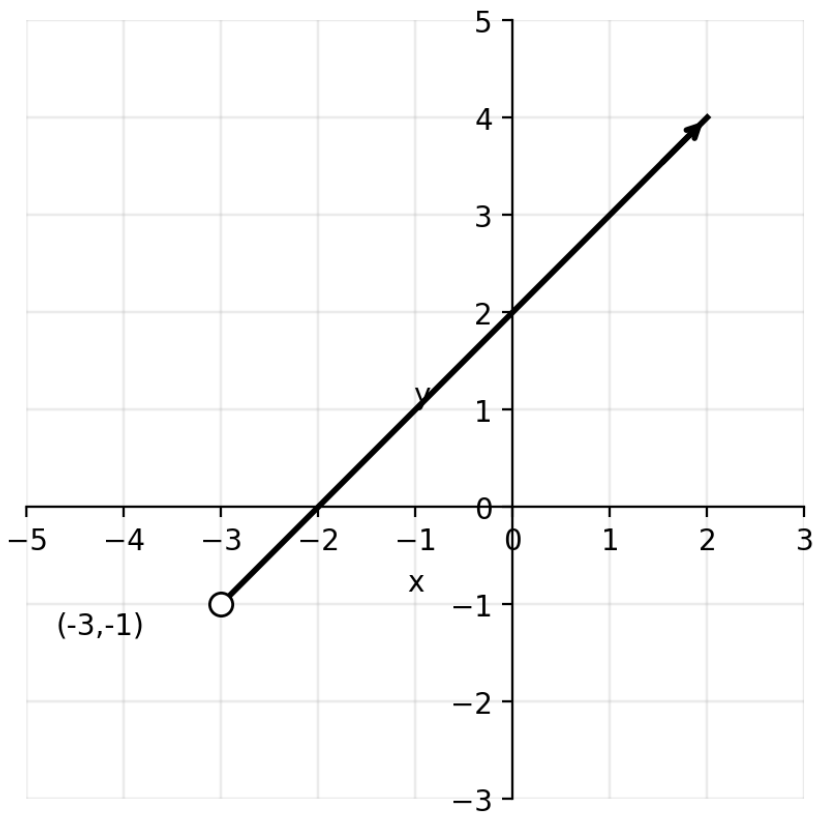
Part A — Practice-Based Questions

Questions 1–15

These questions are based on the attached practice test, with changed numbers and slightly increased difficulty.

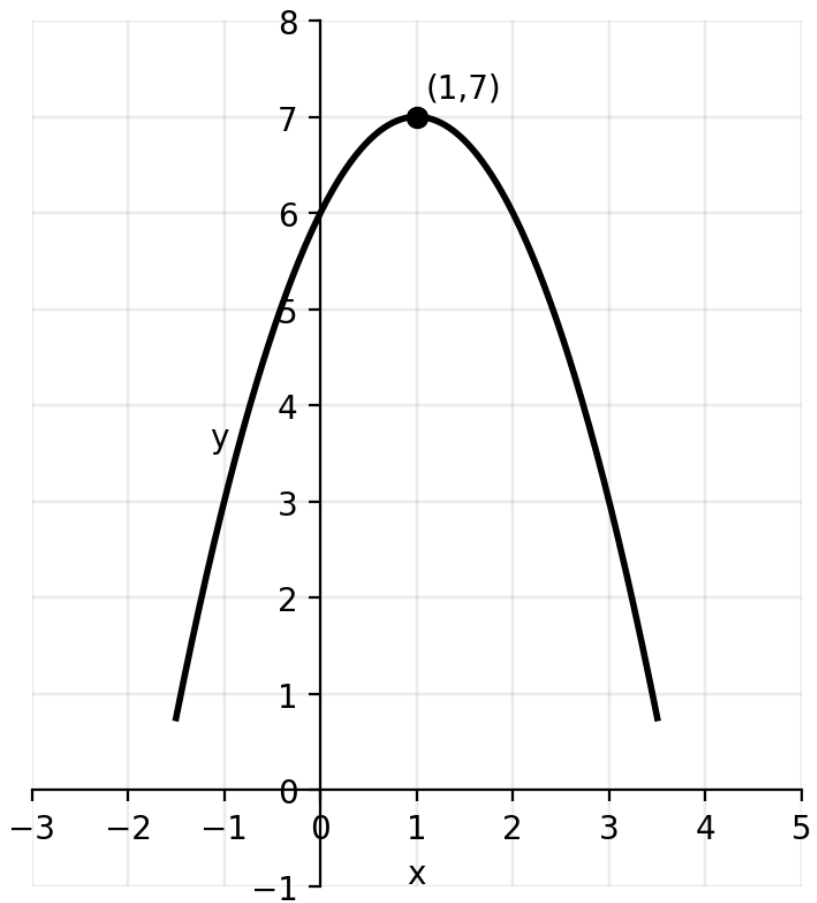
1. Write down the domain and range of each graph.

(a)



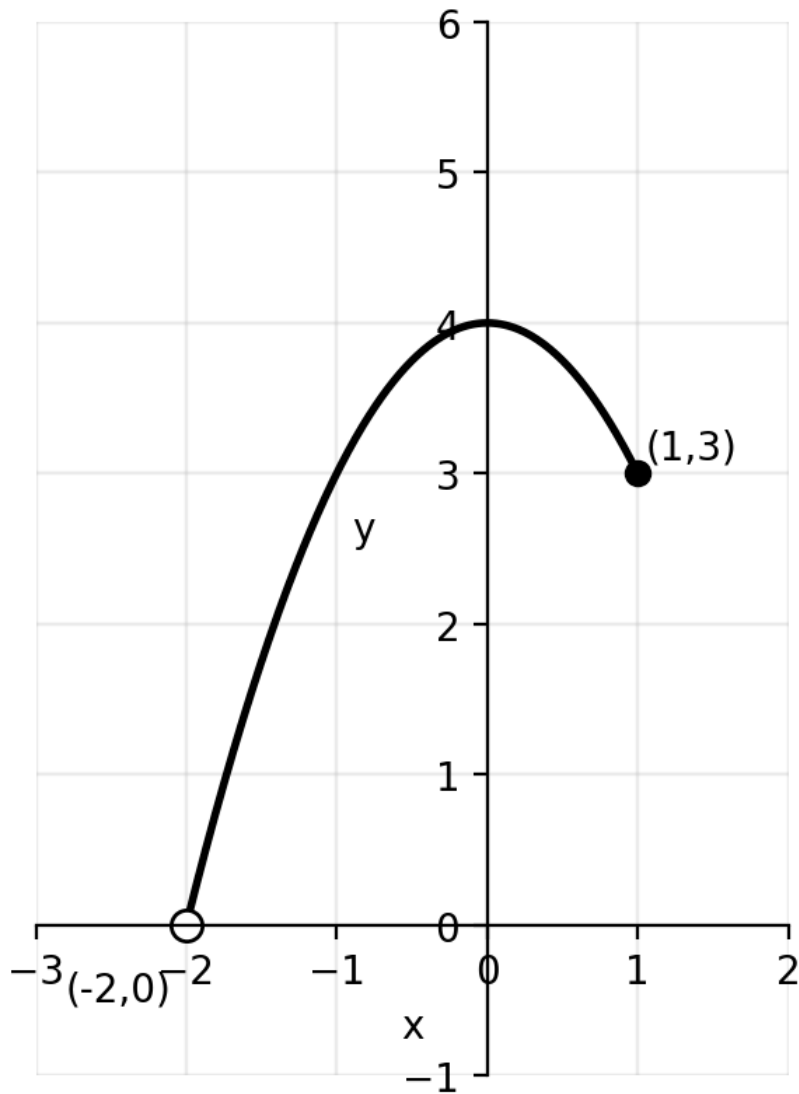
The graph is a ray starting at the open point $(-3, -1)$ and going up to the right.

(b)



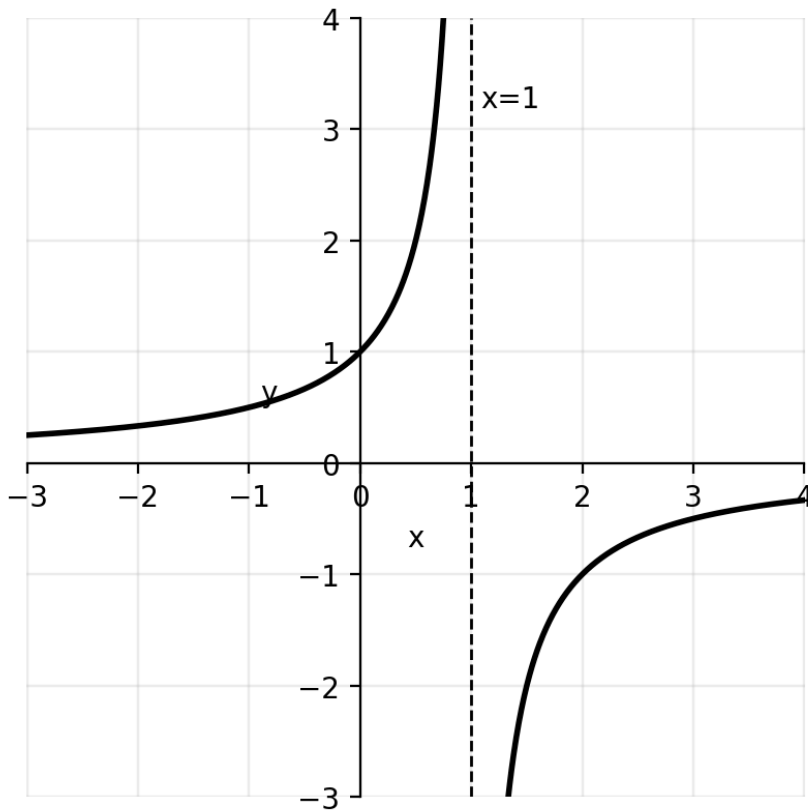
The graph is a downward-opening parabola with maximum point $(1, 7)$.

(c)



The curve begins at the open point $(-2, 0)$ and ends at the closed point $(1, 3)$.

(d)



The graph has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 0$.

2. Find the largest possible domain of each function.

(a) $y = \sqrt{4x + 5} - 2$

(b) $y = \frac{7x - 1}{3 - x}$

(c) $y = \frac{1}{\sqrt{5x - 4}}$

(d) $y = \log_3(2x + 7)$

3. You are given the function

$$f(x) = -x^4 + 2x^2 + 1$$

defined for $0 \leq x \leq 2$.

Write down the range of f .

4. Given $f(x) = 2x$ and $g(x) = 3x - 4$, write down the values of:

(a) $f \circ g(2)$

(b) $g \circ f(3)$

(c) $g \circ g(1)$

(d) $f \circ g \circ f(-2)$

5. Given $f(x) = 4x - 5$ and $g(x) = \frac{x}{4} + 2$, find in simplest form:

(a) $f \circ g(x)$

(b) $g \circ f(x)$

(c) $f \circ f(x)$

6. Given $f(x) = x - 4$ and $g(x) = 5x + 10$, find in the form $ax + b$:

(a) $f^{-1}(x)$

(b) $g^{-1}(x)$

(c) $f^{-1} \circ g^{-1}(x)$

(d) $g^{-1} \circ f^{-1}(x)$

7. Consider the functions

$$f(x) = 2x^2 - 12x + 23 \quad \text{and} \quad g(x) = x + 1$$

(a) Write down the largest possible domain and range of $f(x)$.

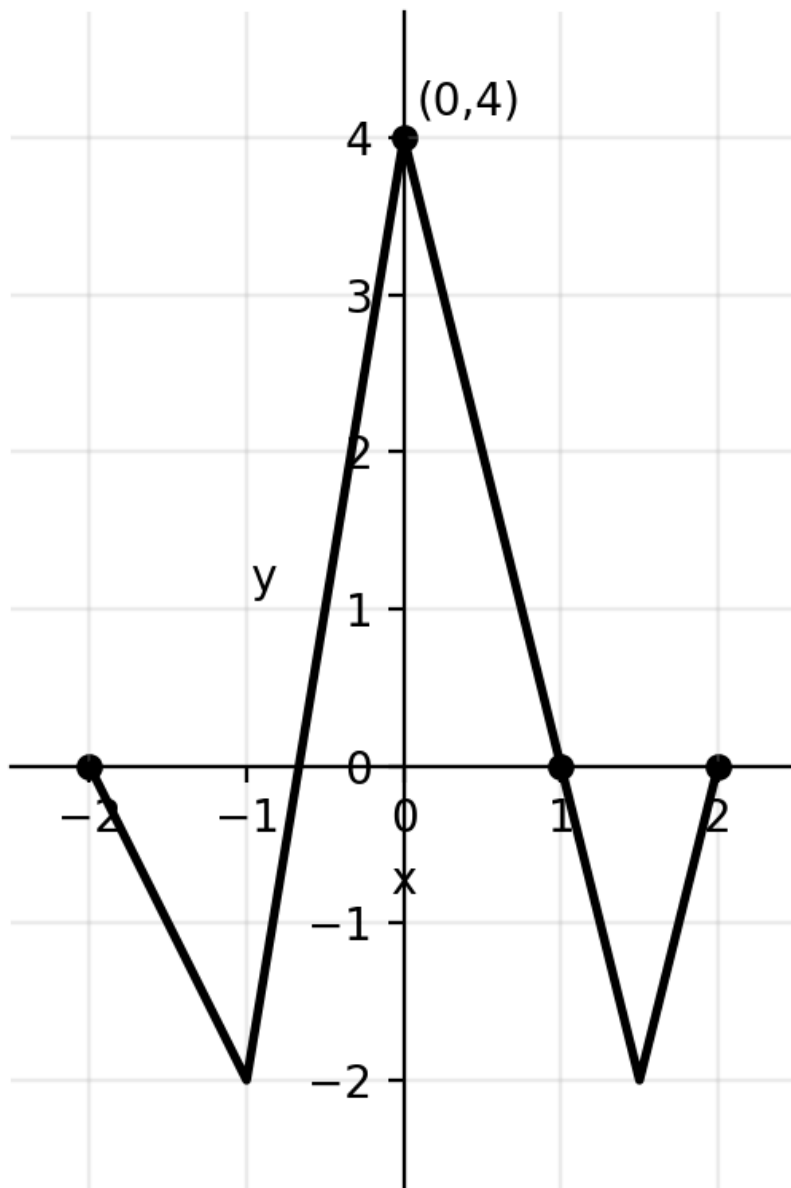
(b) Let $h(x) = f \circ g(x)$. Find an expression for $h(x)$ in the form $ax^2 + bx + c$.

(c) Expand and simplify $2(x - 5)^2 + 1$.

(d) The domain of $h(x)$ is now limited to $x \geq a$ such that this function has an inverse. Write down the smallest possible value of a .

(e) For the value of a found in part (d), find an expression for $h^{-1}(x)$.

8. Let $f(x) = 3x + 1$. The graph of $g(x)$ is shown below.



From the graph, $g(0) = 4$, $g(1) = 0$, and $g(-2) = 0$.

Write down the value of:

(a) $f \circ g(0)$

(b) $\sqrt{g \circ f(-1)}$

9. The function $p(x) = x^2 - 6x + 8$ is defined for $x \geq 3$.

(a) Explain why this restriction allows p to have an inverse.

(b) Find $p^{-1}(x)$.

(c) State the domain of $p^{-1}(x)$.

10. Consider $f(x) = \sqrt{2x - 1}$.

(a) Find the largest possible domain of f .

(b) Write down the range of f .

(c) Explain why f has an inverse.

(d) Find $f^{-1}(x)$.

11. Given $f(x) = x^2 + 2$ and $g(x) = 2x - 1$, find:

(a) $f \circ g(x)$

(b) $g \circ f(x)$

(c) $f \circ g(3)$

(d) $g \circ f(-2)$

12. For the function

$$q(x) = \frac{x+2}{x-3}$$

(a) State the largest possible domain.

(b) Find the value of $q(5)$.

(c) Find $q^{-1}(x)$.

(d) State the value that is not in the range of q .

13. A function is defined by

$$r(x) = \begin{cases} x+2 & \text{for } -3 \leq x < 1 \\ 5-x & \text{for } 1 \leq x \leq 4 \end{cases}$$

Write down:

(a) the domain of r

(b) the range of r

(c) whether r has an inverse on this whole domain, with a reason

14. Solve each equation.

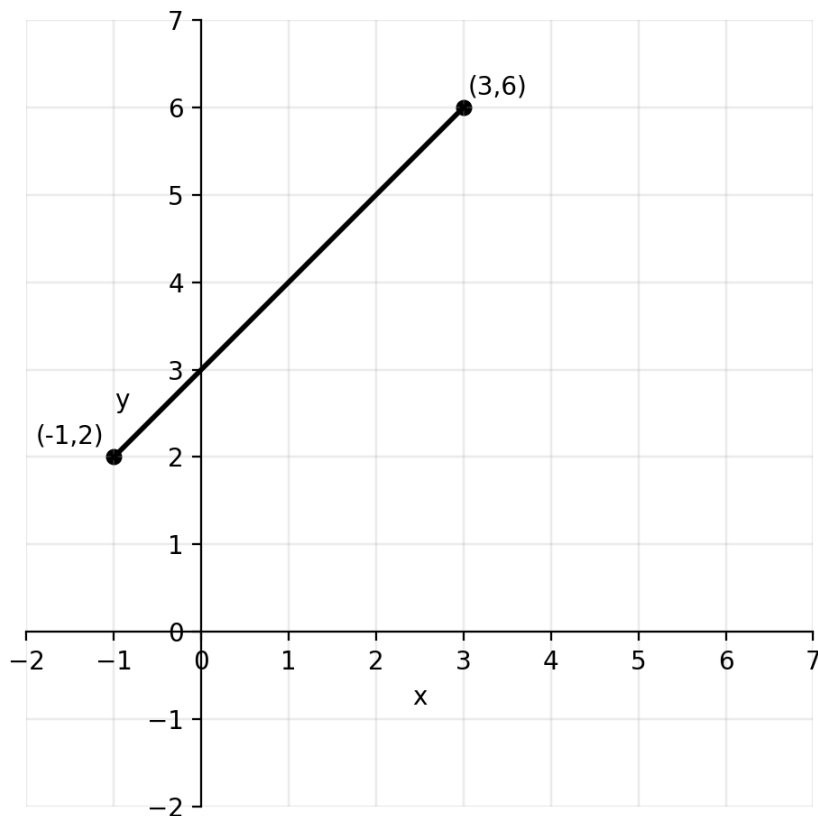
(a) $f(x) = 11$ for $f(x) = 3x - 4$

(b) $f^{-1}(x) = 5$ for $f(x) = 2x + 1$

(c) $f(f(x)) = 19$ for $f(x) = 2x + 3$

15. On the grid below, sketch the graph of $y = f^{-1}(x)$ if the graph of $y = f(x)$ is the line segment from $(-1, 2)$ to $(3, 6)$.

Also state the domain and range of f^{-1} .



Part B — IB Criterion A Unfamiliar Questions

Questions 16–30

These questions are unfamiliar but solvable using the same ideas.

16. A student says:

"If $f \circ g(x) = g \circ f(x)$ for one value of x , then the two composite functions must be equal for all x ."

Use

$$f(x) = 2x + 1 \quad \text{and} \quad g(x) = x^2$$

to decide whether the statement is true or false.

You must:

- find $f \circ g(x)$
- find $g \circ f(x)$
- solve $f \circ g(x) = g \circ f(x)$
- make a conclusion

17. Let

$$f(x) = \sqrt{x+4} \quad \text{and} \quad g(x) = x^2 - 8$$

- (a)** Find $f \circ g(x)$.
- (b)** Find the largest possible domain of $f \circ g$.
- (c)** Explain why the domain is not simply the domain of f .

18. The function

$$h(x) = \frac{3x-2}{x+4}$$

is defined for $x \neq -4$.

- (a)** Find $h^{-1}(x)$.
- (b)** Show that $h \circ h^{-1}(x) = x$.
- (c)** State the value not included in the range of h .

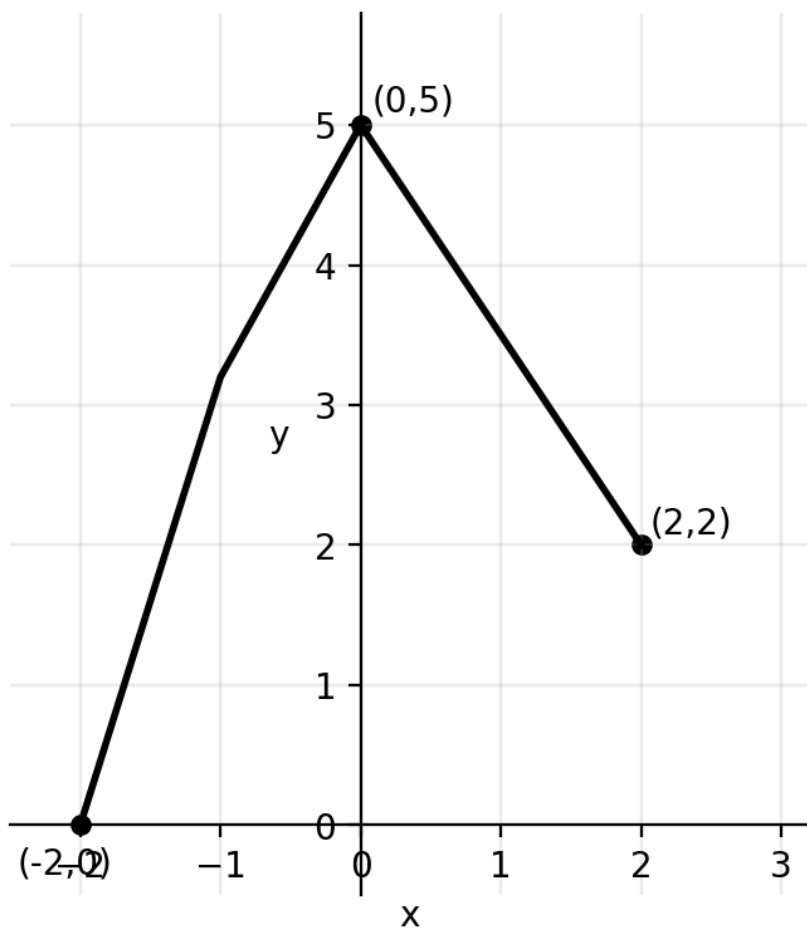
19. The function

$$f(x) = x^2 - 10x + 30$$

is defined for $x \geq 5$.

- (a)** Write $f(x)$ in completed-square form.
- (b)** Find $f^{-1}(x)$.
- (c)** State the domain of $f^{-1}(x)$.
- (d)** Explain what changes in the inverse if the original domain were $x \leq 5$ instead.

20. Consider the graph of $y = g(x)$.



The closed points shown are $(-2, 0)$, $(0, 5)$, and $(2, 2)$.

(a) State the domain and range of g .

(b) Is g one-to-one on the interval shown? Give a reason.

(c) Explain why g^{-1} does or does not exist as a function on this whole domain.

21. Let

$$f(x) = ax + b$$

where a and b are constants.

Given that $f(3) = 11$ and $f^{-1}(7) = 1$, find a and b .

22. A student writes:

$$(f \circ g)^{-1}(x) = f^{-1}(x) \circ g^{-1}(x)$$

Use

$$f(x) = x + 2 \quad \text{and} \quad g(x) = 3x$$

to test this claim.

You must:

- find $f \circ g(x)$
- find $(f \circ g)^{-1}(x)$
- find $f^{-1}(x) \circ g^{-1}(x)$
- state the correct general order for inverses of composite functions

23. Solve for x :

$$f^{-1}(x) = g(x)$$

where

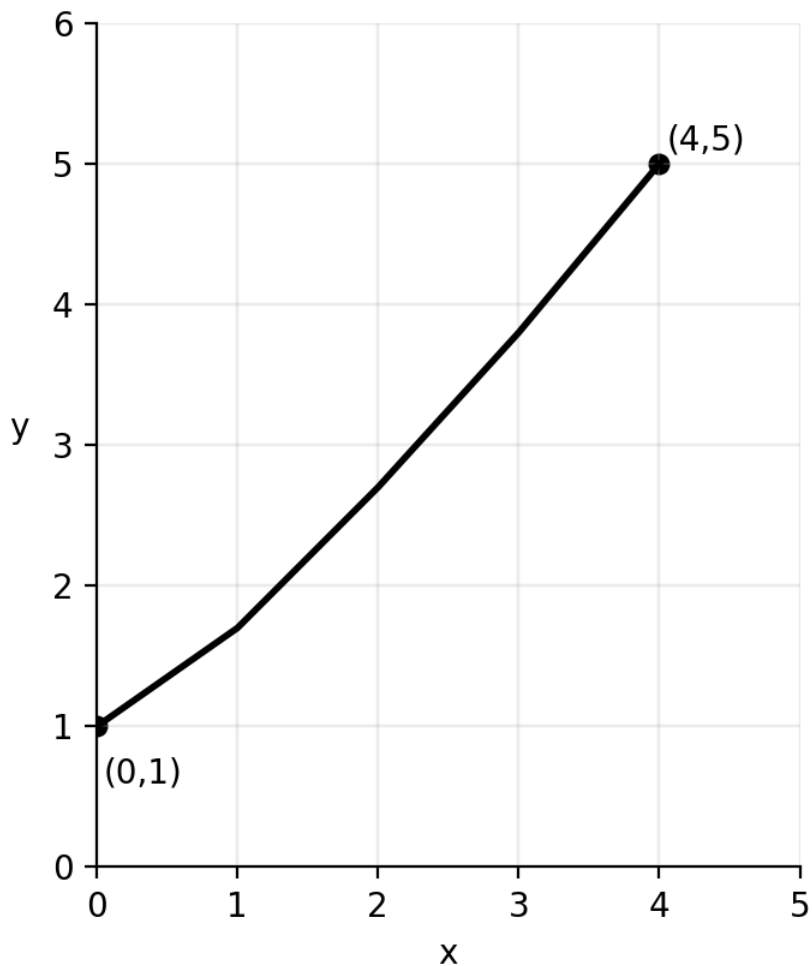
$$f(x) = 2x - 3 \quad \text{and} \quad g(x) = x + 1$$

24. Let

$$m(x) = \frac{1}{\sqrt{x^2 - 16}}$$

- (a) Find the largest possible domain of m .
- (b) Explain why $x \neq \pm 4$ is not a complete domain statement.
- (c) Give one value of x that satisfies $x \neq \pm 4$ but is still not allowed.

25. The graph of $y = f(x)$ is shown below.



The graph is a one-to-one curve from $(0, 1)$ to $(4, 5)$.

- (a) Write down the domain and range of f .
- (b) Write down the coordinates of two points on f^{-1} .
- (c) State the domain and range of f^{-1} .
- (d) Explain how the graph of f^{-1} is related to the graph of f .

26. Let

$$f(x) = 3x - 7$$

Find the value of x such that

$$f^{-1}(x) = f(x)$$

27. Consider

$$f(x) = |x - 2|$$

- (a) Explain why f does not have an inverse on its largest possible domain.

(b) State a restriction on the domain that would allow an inverse to exist.

(c) For your chosen restriction, find the inverse.

28. Let

$$f(x) = x^2 + 4x + 7 \quad \text{and} \quad g(x) = x - 2$$

(a) Find $f \circ g(x)$.

(b) Write your answer in completed-square form.

(c) State the smallest value a such that restricting the domain to $x \geq a$ makes the composite function invertible.

(d) Find the inverse for that restricted domain.

29. A student claims that for

$$f(x) = 4x + 1$$

the inverse is

$$f^{-1}(x) = \frac{1}{4x + 1}$$

(a) Explain why this is incorrect.

(b) Find the correct inverse.

(c) Use a specific input value to check that your inverse works.

30. A machine applies two operations in order:

- first multiply the input by 2
- then subtract 5

A second machine applies:

- first square the input
- then add 1

Let the first machine be f and the second machine be g .

(a) Write expressions for $f(x)$ and $g(x)$.

(b) Find $f \circ g(x)$.

(c) Find $g \circ f(x)$.

(d) Explain in words why these two composite functions are different.

Answer Key

Part A

1.

(a) $D : x > -3, R : y > -1$

(b) $D : x \in \mathbb{R}, R : y \leq 7$

(c) $D : -2 < x \leq 1, R : 0 < y \leq 3$

(d) $D : x \neq 1, R : y \neq 0$

2.

(a)

$$4x + 5 \geq 0 \Rightarrow x \geq -\frac{5}{4}$$

(b)

$$3 - x \neq 0 \Rightarrow x \neq 3$$

(c)

$$5x - 4 > 0 \Rightarrow x > \frac{4}{5}$$

(d)

$$2x + 7 > 0 \Rightarrow x > -\frac{7}{2}$$

3.

$$f(0) = 1, \quad f(2) = -7$$

$$f'(x) = -4x^3 + 4x = 4x(1 - x^2)$$

Critical value in the interval: $x = 1$

$$f(1) = 2$$

So the range is

$$-7 \leq f(x) \leq 2$$

4.

(a) $g(2) = 2$, so $f(g(2)) = f(2) = 4$

(b) $f(3) = 6$, so $g(f(3)) = g(6) = 14$

(c) $g(1) = -1$, then $g(-1) = -7$

(d) $f(-2) = -4$, then $g(-4) = -16$, then $f(-16) = -32$

5.

(a)

$$f \circ g(x) = 4\left(\frac{x}{4} + 2\right) - 5 = x + 3$$

(b)

$$g \circ f(x) = \frac{4x - 5}{4} + 2 = x + \frac{3}{4}$$

(c)

$$f \circ f(x) = 4(4x - 5) - 5 = 16x - 25$$

6.

(a) $f^{-1}(x) = x + 4$

(b) $g^{-1}(x) = \frac{x}{5} - 2$

(c)

$$f^{-1} \circ g^{-1}(x) = \frac{x}{5} + 2$$

(d)

$$g^{-1} \circ f^{-1}(x) = \frac{x - 6}{5}$$

7.

(a)

$$f(x) = 2x^2 - 12x + 23 = 2(x - 3)^2 + 5$$

Largest domain: $x \in \mathbb{R}$

$$R : f(x) \geq 5$$

(b)

$$h(x) = f(x + 1) = 2(x + 1)^2 - 12(x + 1) + 23 = 2x^2 - 8x + 13$$

(c)

$$2(x-5)^2 + 1 = 2x^2 - 20x + 51$$

(d)

$$h(x) = 2x^2 - 8x + 13 = 2(x-2)^2 + 5$$

So $a = 2$.

(e)

$$y = 2(x-2)^2 + 5$$

$$y - 5 = 2(x-2)^2$$

Since $x \geq 2$,

$$x = 2 + \sqrt{\frac{y-5}{2}}$$

Therefore

$$h^{-1}(x) = 2 + \sqrt{\frac{x-5}{2}}$$

8.

(a)

$$f \circ g(0) = f(4) = 13$$

(b)

$$f(-1) = -2, \quad g(-2) = 0, \quad \sqrt{g \circ f(-1)} = 0$$

9.

(a) On $x \geq 3$, the parabola is one-to-one.

(b)

$$p(x) = x^2 - 6x + 8 = (x-3)^2 - 1$$

$$y + 1 = (x-3)^2$$

Since $x \geq 3$,

$$x = 3 + \sqrt{y+1}$$

So

$$p^{-1}(x) = 3 + \sqrt{x+1}$$

(c) Domain of p^{-1} : $x \geq -1$

10.

(a)

$$2x - 1 \geq 0 \Rightarrow x \geq \frac{1}{2}$$

(b) $y \geq 0$

(c) It is increasing on its domain, so it is one-to-one.

(d)

$$y = \sqrt{2x - 1}$$

$$y^2 = 2x - 1$$

$$x = \frac{y^2 + 1}{2}$$

Therefore

$$f^{-1}(x) = \frac{x^2 + 1}{2}$$

11.

(a)

$$f \circ g(x) = (2x - 1)^2 + 2 = 4x^2 - 4x + 3$$

(b)

$$g \circ f(x) = 2(x^2 + 2) - 1 = 2x^2 + 3$$

(c) $g(3) = 5$, so $f(5) = 27$

(d) $f(-2) = 6$, so $g(6) = 11$

12.

(a) $x \neq 3$

(b)

$$q(5) = \frac{5 + 2}{5 - 3} = \frac{7}{2}$$

(c)

$$y = \frac{x + 2}{x - 3}$$

$$y(x - 3) = x + 2$$

$$yx - 3y = x + 2$$

$$x(y - 1) = 3y + 2$$

$$x = \frac{3y + 2}{y - 1}$$

So

$$q^{-1}(x) = \frac{3x + 2}{x - 1}$$

(d) $y \neq 1$

13.

(a) $-3 \leq x \leq 4$

(b) First part gives $-1 \leq y < 3$. Second part gives $1 \leq y \leq 4$. Overall range: $-1 \leq y \leq 4$

(c) No. It is not one-to-one on the whole domain, so it does not have an inverse there.

14.

(a)

$$3x - 4 = 11 \Rightarrow x = 5$$

(b) If $f^{-1}(x) = 5$, then $x = f(5) = 11$

(c)

$$f(f(x)) = 2(2x + 3) + 3 = 4x + 9$$

$$4x + 9 = 19 \Rightarrow x = \frac{5}{2}$$

15. Endpoints swap:

$$(-1, 2) \mapsto (2, -1), \quad (3, 6) \mapsto (6, 3)$$

So $y = f^{-1}(x)$ is the line segment from $(2, -1)$ to $(6, 3)$.

Domain of f^{-1} : $2 \leq x \leq 6$

Range of f^{-1} : $-1 \leq y \leq 3$

Part B

16.

$$f \circ g(x) = 2x^2 + 1$$

$$g \circ f(x) = (2x + 1)^2 = 4x^2 + 4x + 1$$

Set them equal:

$$2x^2 + 1 = 4x^2 + 4x + 1$$

$$0 = 2x^2 + 4x = 2x(x + 2)$$

$$x = 0 \text{ or } x = -2$$

They are equal for some values, but not for all values. The statement is false.

17.

(a)

$$f \circ g(x) = \sqrt{(x^2 - 8) + 4} = \sqrt{x^2 - 4}$$

(b)

$$x^2 - 4 \geq 0$$

$$x \leq -2 \quad \text{or} \quad x \geq 2$$

(c) The output of g becomes the input of f , so the restriction must be applied after substitution.

18.

(a)

$$y = \frac{3x - 2}{x + 4}$$

$$yx + 4y = 3x - 2$$

$$x(y - 3) = -(2 + 4y)$$

$$x = \frac{2 + 4y}{3 - y}$$

So

$$h^{-1}(x) = \frac{2 + 4x}{3 - x}$$

(b) Substituting $h^{-1}(x)$ into h simplifies to x .

(c) The range excludes $y = 3$.

19.

(a)

$$f(x) = x^2 - 10x + 30 = (x - 5)^2 + 5$$

(b)

$$y = (x - 5)^2 + 5$$

Since $x \geq 5$,

$$x = 5 + \sqrt{y - 5}$$

So

$$f^{-1}(x) = 5 + \sqrt{x - 5}$$

(c) Domain of f^{-1} : $x \geq 5$

(d) If the original domain were $x \leq 5$, the inverse would be

$$f^{-1}(x) = 5 - \sqrt{x - 5}$$

20.

(a) Domain: $-2 \leq x \leq 2$, Range: $0 \leq y \leq 5$

(b) No. Some horizontal lines meet the graph more than once.

(c) Therefore g^{-1} does not exist as a function on this whole domain.

21. From $f(3) = 11$:

$$3a + b = 11$$

From $f^{-1}(7) = 1$, we know $f(1) = 7$:

$$a + b = 7$$

Subtracting gives

$$2a = 4 \Rightarrow a = 2$$

Then

$$b = 5$$

22.

$$f \circ g(x) = f(3x) = 3x + 2$$

So

$$(f \circ g)^{-1}(x) = \frac{x - 2}{3}$$

Also

$$f^{-1}(x) = x - 2, \quad g^{-1}(x) = \frac{x}{3}$$

Then

$$f^{-1} \circ g^{-1}(x) = \frac{x}{3} - 2$$

These are not equal, so the claim is false.

Correct general rule:

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

23. First,

$$f^{-1}(x) = \frac{x+3}{2}$$

Now solve

$$\frac{x+3}{2} = x+1$$

$$x+3 = 2x+2$$

$$x = 1$$

24.

(a)

$$x^2 - 16 > 0$$

$$x < -4 \quad \text{or} \quad x > 4$$

(b) The statement $x \neq \pm 4$ excludes only the boundary values, but not values making the expression inside the square root negative.

(c) One example is $x = 0$.

25.

(a) Domain: $0 \leq x \leq 4$, Range: $1 \leq y \leq 5$

(b) Two points on f^{-1} are $(1, 0)$ and $(5, 4)$

(c) Domain of f^{-1} : $1 \leq x \leq 5$, Range of f^{-1} : $0 \leq y \leq 4$

(d) The graph of f^{-1} is the reflection of the graph of f in the line $y = x$.

26.

$$f^{-1}(x) = \frac{x+7}{3}$$

Set equal:

$$3x - 7 = \frac{x + 7}{3}$$

$$9x - 21 = x + 7$$

$$8x = 28$$

$$x = \frac{7}{2}$$

27.

(a) It is not one-to-one on all real numbers.

(b) One valid restriction is $x \geq 2$.

(c) For $x \geq 2$,

$$f(x) = x - 2$$

so

$$f^{-1}(x) = x + 2$$

with domain $x \geq 0$.

28.

(a)

$$f \circ g(x) = f(x - 2) = (x - 2)^2 + 4(x - 2) + 7 = x^2 + 3$$

(b)

$$x^2 + 3 = (x - 0)^2 + 3$$

(c) The smallest value is $a = 0$.

(d)

$$y = x^2 + 3$$

For $x \geq 0$,

$$x = \sqrt{y - 3}$$

So

$$(f \circ g)^{-1}(x) = \sqrt{x - 3}$$

29.

(a) $\frac{1}{4x + 1}$ is the reciprocal of $f(x)$, not the inverse function.

(b)

$$f^{-1}(x) = \frac{x-1}{4}$$

(c) For example, $f(3) = 13$ and $f^{-1}(13) = 3$.

30.

(a)

$$f(x) = 2x - 5, \quad g(x) = x^2 + 1$$

(b)

$$f \circ g(x) = 2(x^2 + 1) - 5 = 2x^2 - 3$$

(c)

$$g \circ f(x) = (2x - 5)^2 + 1 = 4x^2 - 20x + 26$$

(d) Composition depends on order, so changing the order changes the output expression.

End of test.